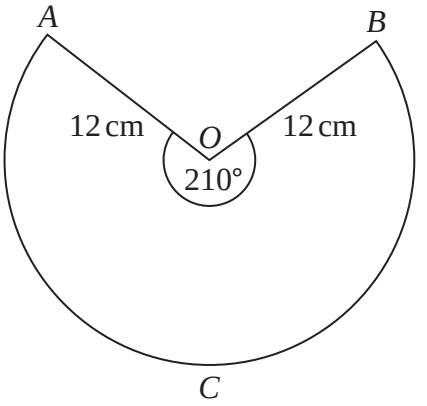
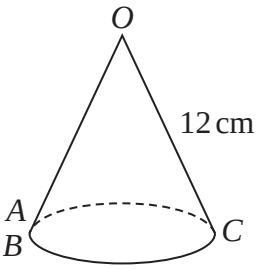
16 The diagram shows sector *OACB* of a circle with centre *O* and radius 12 cm.

Reflex angle *AOB* = 210°

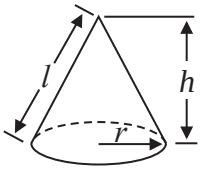
The diagram also shows a hollow cone with vertex *O*.
The curved surface of the cone is formed by joining *OA* to *OB*.
Arc *ACB* forms the circumference of the base of the cone.





Curved surface area of cone = πrl

Volume of cone = $\frac{1}{3}\pi r^2 h$



Calculate the volume of the cone.
Give your answer correct to 3 significant figures.

Sector arc = (210/360) ×2π ×12 = 14π cm

Cone base circumference = 2πr = 14π

1 mark
Arc of the sector becomes the full base circumference.

r = 7 cm; slant height l = 12 cm

Vertical height h = √(12² −7²) = √95 cm

1 mark
Pythagoras in the cone: height is not the slant height.

Volume = (1/3)πr²h = (1/3)π ×7² ×√95

= 500.134139... cm³ = 500 cm³ (3 s.f.)

2 marks total
Use one third πr²h; retain √95 until the final calculation.

500 ..... cm³

(Total for Question 16 is 4 marks)

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